Unit: 6 Unit 4: Absolute Value **Name:** ________________
Topic: Absolute Value Inequalities and Their Applications **Date:** ___________

Advanced (Carolina Reaper)

Q1. Solve $|x-5|<2$.
(A) $3<x<7$
(B) $3leqxleq7$
(C) $1<x<9$
(D) $1leqxleq9$
(E) $0<x<10$

Q2. Graph the solution set of $|x+2|geq4$ on a number line.
(A) $xleq-6$ or $xgeq2$
(B) $x<-6$ or $x>2$
(C) $xleq-2$ or $xgeq6$
(D) $x<-2$ or $x>6$
(E) $xleq2$ or $xgeq6$

Q3. Express the solution set of $|3x-1|>5$ in interval notation.
(A) $(-infty,-$
rac43) cup (
rac63, infty)
(A) $(-infty,-$
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rac43, infty)
(A) $(-infty,-2)cup(2,infty)$
(B) $(-2,2)$
(C) $(-$
rac43,
rac63)

Q4. A bakery sells a total of 240 loaves of bread per day. They pack the bread in bags that hold 12 loaves each. If x is the number of bags sold, then $|x-20|leq2$ represents the possible number of bags sold. What is the range of the total number of loaves sold?
(A) 192 to 288
(B) 216 to 264
(C) 204 to 276
(D) 198 to 282
(E) 210 to 270

Q5. A person spends 2 hours more than the time they sleep. If x is the number of hours slept, then the time spent awake is given by $|x-10|+2$. If the time spent awake is at most 16 hours, what is the range of values for x?
(A) $6leqxleq12$
(B) $4leqxleq14$
(C) $2leqxleq16$
(D) $8leqxleq10$
(E) $0leqxleq8$

Q6. Solve $|2x+5|<11$.
(A) $-7<x<3$
(B) $-8<x<2$
(C) $-6<x<4$
(D) $-4<x<6$
(E) $-3<x<5$

Q7. A group of friends want to buy tickets to a concert. The cost of a ticket is $|x-50|+20$, where x is the number of tickets bought. If the total cost is at least 100 and at most 200, what is the range of the number of tickets bought?
(A) $20leqxleq80$
(B) $30leqxleq70$
(C) $10leqxleq90$
(D) $25leqxleq75$
(E) $40leqxleq60$

Q8. Solve $|x^2-4|>12$.
(A) $x<-4$ or $x>4$
(B) $x<-2$ or $x>6$
(C) $x<-6$ or $x>2$
(D) $x<2$ or $x>6$
(E) $x<-2sqrt3$ or $x>2sqrt3$

Open Ended Questions (FRQ)

Q9. A water tank can hold 1200 gallons of water. Due to a leak, the tank is losing water at a rate of $|2x-10|$ gallons per hour, where x is the number of hours passed. If the tank is currently full, what is the range of values for x such that the tank has at least 600 gallons of water? Provide your answer in interval notation.

Q10. A company produces x units of a product per day. The cost of producing x units is given by $|x-200|+500$. If the company can produce at most 300 units per day and the cost of producing x units is at most

$$1200$$

, what is the range of values for x? Provide your answer in interval notation.

Chef's Tips

- Tip 1: Emphasize the importance of understanding the properties of absolute value inequalities.
- Tip 2: Encourage students to use number lines to visualize the solution sets of absolute value inequalities.
- Tip 3: Remind students to check their work and consider the context of the problem when solving absolute value inequalities.